PROJECT PROPOSAL

Dhiraj Ram

Contents

Business Context 2
- Producers Information 2
- Products Information 2
- Ordering Products 2
- Manage Account and Order History 2
- Producers Dashboard 2
- Order Management 2
- Loyalty Scheme 2
- Accessibility Features 3

Project Aim 3

Functional Requirements and User Acceptance Criteria 3
- Users' Personas 3
- Functional Requirements Table 3

Non-functional Requirements 9
- Performance 9
- Security 10
- Capacity 11
- Scalability 11
- Usability and Accessibility 12

Risk Management 12
- General Project Risks (GPR) 12
- General Technical Risks (GTR) 12
- Project-Based Risks (PBR) 13
- Risk Matrix 13
- Risk Probability and Impact 13
- Prioritised Risks Mitigation 13

Legal Considerations 14

UI Considerations 15

Business Context

Producers Information

This will allow to provide information about the producers, their methods and the benefits of buying locally. Other competitors of the client also have this, which means if the client also has this in their solution, it will be regarded as professional and lead to users being more confident in using the solution. This results in a higher potential of customers buying the client's products.

Products Information

This allows customers to view products with transparent pricing and availability. This will benefit the client because their customers are going to trust them more as the client cares about providing transparent pricing. Furthermore, showing availability to customers of products will lead to less confusion if the products are available or not. This will result in the client providing an enhanced customer experience, which encourages the customers to use the client's services again.

Ordering Products

Allow customer to place orders for collection or delivery. This is industry standard in the digital solutions that the client's competitors have. This means it will make the client's solution up to par with its competitors and lead to trust from customers as they would have seen similar digital solutions. More importantly this provides an easy way for the customers to get their products which means the customers are more likely to buy the client's products and lead to growth of the client's business.

Manage Account and Order History

Allow customers to manage their accounts and view order history. This will benefit the client as customers will find using the solution more convenient, as they do not need to retype their information to buy a product, because it has been stored in their account. Furthermore, providing order history to the customers will make them prone to reordering the same product they bought before and give the client more reoccurring revenue.

Producers Dashboard

Provide a dashboard to enable producers to manage stock levels, update product details and view orders. This will benefit the client as they will be able to provide up-to-date information about their products to their customers. Furthermore, it will make it easy for the client to manage products as they do not have to use external tools to get details on their products and update the products on the digital solution accordingly. This will also save time and money for the client as they do not have to dedicate staff for product management, and it can be done by the producers themselves.

Order Management

This will allow to provide order management, scheduling and tracking features. This will benefit the client as they will be able to provide updates on their customers' orders, which will make the customers less stressed as they do not have to worry about when their order is arriving. This will result in a better customer experience which makes the customer more likely to buy from the client.

Loyalty Scheme

A loyalty scheme that provides discounts and specials offers. This will benefit the client as they will be able to create more regular customers since the customers will be incentivised to buy more from the client. Furthermore, this will give the client a competitive advantage, since other competitors do

not provide this feature, which means customers will more likely want to use the client's services than the competitors' services.

Accessibility Features

Accessibility features to support a wide range of users. This will benefit the client as they will be able to increase the number of potential customers who can use the solution, since the solution can adapt to different needs of the customers to use the solution. Furthermore, including this in the solution will increase the reputation of the client, since customers will think the client cares about providing a good user experience for them.

Project Aim

The overall aim of the project is to provide Greenfield Local Hub (GLH) with an online web platform that has:

- Provide information about the producers, their methods and the benefits of buying locally
- Allow customers to view products with transparent pricing and availability
- Allow customer to place orders for collection or delivery
- Allow customers to manage their accounts and view order history
- Provide a dashboard to enable producers to manage stock levels, update product details and view orders
- Order management, scheduling and tracking features
- A loyalty scheme that provides discounts and specials offers
- Accessibility features to support a wide range of users

Functional Requirements and User Acceptance Criteria

Users' Personas

Bob – a regular customer of the client

Greg – a new customer who's looking to buy food locally

Sophie – a farmer who sells her products through the client

Functional Requirements Table

Functional Requirement	**User Story**	**User Acceptance Criteria**	**Justification**
Users able to register	As Bob, I want to be able to register, so that I can reorder products easily. As Greg, I want to be able to register, so that I can save my order information. As Sophie, I want to be able to register, so that I can sell my products.	• View register button • Navigate to register page on register button interaction, located in the navigation bar • Enter register information • Interact with create button to create an account and be	This feature will allow users to access more features of the application, since data is needed to create a tailored experience for the user, which improves user experience.

		redirected to the log in page	
Users able to log in	As Bob, Greg and Sophie, I want to be able log in to my account, so that I can access the features of the apps using my account data.	• View login button • Navigate to login page on login button interaction, located in the navigation bar • Enter login information • Interact with login button to login into the user's account and be redirected to the home page	This feature will allow users to access their account that already have made, so they do not have to make a new account every time they use the application which improves user experience.
Users able to change their log in details	As Bob, Greg and Sophie, I want to be able to change my log in details, so that I can set new login details.	• View log in button • Navigate to log in page on log in button interaction, located in the navigation bar • View log in change button • Interact with login change button to change login information • Enter new login information • Interact with confirm button to confirm login information	This feature will allow users to update their log in details if they want to change them something more appropriate for them and this would avoid user dissatisfaction, if they used login details that they could not remember or were too long.
Users able to view privacy information	As Bob, Greg and Sophie, I want to be able to view privacy information, so that I know how my data is being used.	• View privacy button • Navigate to privacy page on privacy button interaction, located in the navigation bar • View privacy information • The page must comply with the Web Accessibility Guidelines (WCAG) e.g. keyboard navigation, adjustable text size, high contrast mode.	This feature will ensure that the client is compliant with GDPR laws by being transparent about how user data is used by the application.
Users able to view information about the producers, their methods and	As Bob, I want to be able to view information about the producers, their methods, so that I know who produces	• View home page • View "Our Producers" page	This will allow customers to understand who the client works with, how the products

the benefits of buying locally	my food and how it is produced. As Greg, I want to be able to view information about the producers, their methods and the benefits of buying locally, so that I know if I should buy the products from this business or not. As Sophie, I want to be able to view information about the producers, so that I know if I am working with a business who can sell my product properly.	• View "Our Producers" navigation button on the navigation bar • View producers' section • View methods' section • View "Benefits of buying locally" section • Navigate to producers' page on "Our Producers" button interaction, located in the producers' section and the navigation bar	are produced and why they should buy locally. This will build trust with customers as they know who makes their food and how. This will benefit the client with customers likely to buy from them as the customers knows the information about the business.
Customers able to view products with transparent pricing and availability	As Bob and Greg, I want to be able to view products with transparent pricing and availability, so that I if the prices have changed or if a product is no longer available.	• View products page • View product cards • View product details • View products button on the navigation bar	This will ensure customers know the different products that are offered and their prices. This will allow the client to show all their products to the customers and increase the chance of a purchase being made by the customers.
Customers able to place orders for collection or delivery	As Bob and Greg, I want to be able to place orders for collection or delivery, so that I can choose how I want my products to be received.	• View products page • View order page • View basket page • View product cards • View product details • View products button on the navigation bar • View basket button on the navigation bar • View "Add" button on product card • View "Add" button on a specific product's page • Product added to basket when interacted with "Add" button on product card	This will allow customers to place orders online which means they do not need come to food market to purchase products they want to buy. This will lead to better customer satisfaction which encourage those customers to recommend the client to other potential customers through word of mouth.

		• Navigate to a specific product's page when clicked on the product card • Interact with basket button on the navigation bar to view basket items • View checkout button on basket side bar • Interact with checkout button to navigate to order page • View order details • View order form fields • Interact with order form fields to fill the order details out • View "Order" button • Interact with "Order" button to order the items in the basket	
Customers able to view order history	As Bob, I want to be able to view order history, so that I know what I have ordered before form the business or to reorder products from the order history.	• View orders page • View orders history section • View orders card on the orders history section • View basic order details on orders card on the orders history section • Navigate to orders history section after interacting with order card on orders history section	This will allow customers view what they have order before with client and increase the likelihood of the customer reordering the products. This will benefit the client as they will be able to have regular customers and which results in stable revenue for the client.
Provide a dashboard for producers	As Sophie, I want to be able to use a dashboard for product management, so that I manage stock levels, update product details and view orders.	• View dashboard page • View dashboard button on navigation bar • Navigate to dashboard page after interacting with dashboard button on navigation bar • View products and stock levels on dashboard	This will allow producers to manage their products that they sell through the client. This will allow the client to provide up to date product information to customers, which means a greater chance of customers

		• View buttons on products in the dashboard to edit details • Navigate to edit products page when clicked product card on dashboard • View product details fields • Interact with product details fields to change the details of a product • Interact with submit button to confirm changes to the product details.	buying products from the client.
Provide order management, scheduling and tracking features	As Bob and Greg, I want to be able to do order management, scheduling and use tracking features, so that I know when my order is going to be delivered to me. As Sophie, I want to be able to do order management, scheduling and use tracking features, so that I can keep the customers updated about their orders status.	• View order history page • View home page • View order history section on home page • View "All Orders" heading on order history section on home page • Interact with "All Orders" heading to be navigate to the order history page • View order cards on order history page • View order details such as status on order card on order history page • View deny and approve buttons on order cards as a "Admin" or "Producer" user • Interact with deny and approve buttons on cards as a "Admin" or "Producer" user	This will allow producers to update customers on their orders. And customers will be able to manage their orders. This will benefit the client as the customers will not be frustrated if they do not know their order status. This will also benefit the client in terms of saving money hiring staff to manage order status, since they will not need staff to do this task because it will be done by the application.
A loyalty scheme that provides discounts and specials offers	As Bob, I want to be able to get discounts and special offers, so that I can get	All users: • View products page • View register page • View log in page	This will allow to create a community of loyal customers as their being rewarded

	rewarded for my loyalty with the business. As Sophie, I want to be able to provide discounts and specials offers, so that I can encourage customers to buy my products more often.	• View product cards on products page • View product basic details on product cards • View product all details on product specific page • View discounts on product prices on product specific page Producer users: • View discount type field on product specific page • Interact with discount type field on product specific page to change discount type only if the product is of the producer • Interact view "Save Changes" to save the changes made to the discount type field	for their contribution to the business. Furthermore, having this rewards system will entice new customers to become frequent customers, so they too can get discounted products. This ultimately will allow the client's business to grow more quickly.
Users able to change the application according to their accessibility needs	As Bob, Greg and Sophie, I want to be able to change the application according to my accessibility needs, so that I can use the application comfortably.	• View navigation bar • View theme switch button on navigation bar • View text-to-speech button on navigation bar • View high contrast button on navigation bar • View font size adjustment buttons on navigation bar • Interact with theme switch button to change between light mode and dark mode • Interact with text-to-speech button to use text-to-speech • Interact with font size adjustment buttons to increase or decrease font size	This will allow users to change the application according to their accessibility needs. For example, text-to-speech, high contrast more, or font size adjusting. This benefit the client as they will have different ages and different ability of customers, and by providing this they will be able to serve them and increase the potential customers they can sell products to.

Add access control pages based on user roles	As Bob, Greg and Sophie, I want to be able view relevant pages to me, so that I cannot do something that I am not allowed to do.	• View all pages • View access denied page • Access denied page displayed when unauthorised user accesses a page • Pages shown to the authorised user roles • Actions not processed for unauthorised roles on unauthorised pages	This will allow users to only have access to pages that they have given permissions to. This will benefit the client as the application is going to be more secure and reduce the chance of a data breach, and that leading the client to pay a fine.

Non-functional Requirements

Performance

Requirement	**Justification**	**KPI**
The webpages must load in industry standard time.	This will allow a good user experience if webpages are loaded quickly and create good reputation for the client's website.	Webpages containing just text should load within 15 milliseconds. Webpages containing less than 20 high quality images should load within 30 milliseconds. Webpages containing more than 30 high quality images should load within 50 milliseconds.
The system must handle a specific number of requests without degradation.	This will ensure the core function of the website will still work at a busy day without slowing down website significantly.	On average the system should handle 30 requests per second without degradation.
CPU usage should not exceed a set amount	This will ensure low end devices can use the website without sacrificing significant speed of the website.	CPU usage should be under 20% on a S20 phone. CPU usage should be under 15% on iPhone15 Pro Max.
System should operate correctly under a multiplied expected load.	This will allow the website to still work on days when there is more load than expected, which means the client won't lose many customers due to systems not working.	System should operate correctly under 4 times expected load.
Input delay should be at or under industry standards	This will ensure the website feels responsive which will create a better user experience over other competitors who have less responsive websites.	Input delay should be at or under 200 milliseconds when interacting with functions that process more than 100kb of data.

		Input delay should be at or under 100 milliseconds when interacting with functions that process more than 10kb of data.
Payment should be processed quickly.	This will allow the customers to feel like that the website is responsive as it will not take long to process the payment.	The payment processing speed should be at an industry standard.
Orders' status should be updated fast when changed.	This will allow the customers to view their orders fast and not get frustrated since they were not able to view the orders' status change fast.	Orders' status being updated under 2 minutes after the status has been changed, during 50 or more orders' statuses are being changed.

Security

Requirement	**Justification**	**KPI**
All data should be secured using industry standard encryption.	This will allow the users data to be safe in case of a data breach and limit any fines to the client.	Data should be saved using SHA256 encryption.
Account password should be considered complex by industry standards	This will reduce chances of someone guessing the password through brute force attacks and add extra layer of protection over hashed the password.	Account password should be at least 8 character long, contain letters, numbers and special characters to be consider complex.
Data should be encrypted during transit.	This will allow the data to be protected from man in the middle attacks and give the users greater confidence in the website's data protection.	Website should enforce HTTPS on data when it is in transits. Data being transited should not contain sensitive information that is not encrypted before being transited.
There should be explicit user consent for cookies	This will allow the client to adhere to GDPR laws and protect themselves from any fines considering cookies.	Cookies should be visible on any page of the website with a clear way to accept or deny the cookies.
Account details of the user should be censored.	This will ensure others cannot access details of the user, by viewing the user's screen.	Important details like passwords and emails will be replaced with asterisks or dots, these will be the same amount regardless the number of characters.
Payment details should not be shown on the screen by default.	This will ensure the customers' details will not be stolen by someone viewing their screen when they are entering their payment details.	Card numbers will be replaced with asterisks or dots, when they are being typed in or just being displayed.

		Card holder's full name should be abbreviated to only show the surname on screen.
Users given specific roles	This will ensure the users that are not supposed have access to certain pages will not have to access to it.	Users are only able to view pages that they are allowed based on their roles.

Capacity

Requirement	Justification	KPI
The website should be able to handle a max number of users without the system crashing	This will ensure the website can still be available on high traffic days.	The website should be able to handle max of 1000 users at the same time without the system crashing.
The website should be able to handle many transactions per second under peak load	This will allow the business to continue during high traffic events such as promotions.	500 transactions per second should be sustained with less than or equal to 5% error rate.
The website should be able to store data of many customers without overwriting previous data of other customers.	This will allow the website to handle the storage customers' details and orders, without cost the constant need to upgrade the storage. Additionally, it will avoid customers getting frustrated if they website cannot store their details or orders.	The website should be able to store up to 10TB of data for 10,000 customer accounts.

Scalability

Requirement	Justification	KPI
New functions should be added at an industry standard pace to the application	This will allow the client to show that they care about their customers user experience using the website, which will make the client seem more trustworthy.	On average, every new major function should be added within 5 to 10 days. On average, every new minor function should be added within 1 to 3 days.
Traffic should be dispersed across different servers.	This will ensure that website will still be providing a smooth experience even if there is an increase in users.	On average traffic should be dispersed evenly across all the different servers.
The code should be easily maintained and understandable by others.	This will allow the website to get more features more efficiently as other developers can understand the code easily.	A third party should be able to understand the coding conventions and logic being used in the solution and edit the code easily after a week of reviewing.

Usability and Accessibility

Requirement	Justification	KPI
Webpage should adapt to different device sizes	This will allow the website to be used from many different websites which means more potential users of the website and allows the client to follow WCAG for their website.	Webpage elements all are visible at different device sizes.
Website should have intuitive and consistent navigation	This ensures user can easily find the right content and reduces frustration, which reduce drop-off rates of the website.	Any page should be able to be reached within 3 clicks from the homepage.
Interactive elements should provide immediate visual feedback when interacted with	Ensures users are not confused since they will be able to tell that their actions are recognised by the website. This leads to better user experience.	All interactive components should change their state (colour, animation, size and etc) within 1 second of user interaction.
Should have maintain consistent visual design and content tone throughout all pages	This will allow for better brand recognition and build trust for customers, due to consistency in the visual design.	All pages should fully adhere to the design document of the website.
The solution must show errors in an informative and understandable way.	This will ensure users do not get frustrated, if they have errors in the input or output of the solution. Additionally, this will allow the website to be professional, as it is to industry standard to show error in an informative and understandable manner.	Users should not have no complaints of them being confused or frustrated with errors.

Risk Management

General Project Risks (GPR)

1. Scope creep
2. Schedule delays
3. Quality issues
4. Technical failure

General Technical Risks (GTR)

1. DDOS Attack
2. Broken functionality after updates
3. Data loss or corruption
4. Scalability limitation under high traffic
5. Third party service failure
6. Server outages

Project-Based Risks (PBR)

1. Ordering system errors
2. Back-end access by unauthorised users
3. Payment provider not working
4. Producers adding wrong product details
5. Incorrect stock and availability
6. Producer struggling to adopt using the dashboard

Risk Matrix

Key:

- Orange Coloured Cells = Somewhat high priority risks
- Red Coloured Cells = High priority risks

Impact	High	GTR1, GTR3	GPR3, GTR6, PBR2	GPR1, GPR2, GTR4, PBR1, PBR3, PBR5	PBR6
	Significant		GTR2, GPR4, GTR5		PBR4
	Moderate				
	Low				
		Low	Moderate	Significant	High
		Probability			

Risk Probability and Impact

Risk Description	**Probability**	**Impact**
DDOS Attack	Low	High
Broken functionality after updates	Moderate	Significant
Scope creep	Significant	High
Schedule delays	Significant	High
Quality issues	Moderate	High
Technical failure	Moderate	Significant
Data loss or corruption	Low	High
Scalability limitation under high traffic	Significant	High
Third-party service failure	Moderate	Significant
Server outages	Moderate	High
Ordering system errors	Significant	High
Back-end access by unauthorised users	Moderate	High
Payment provider not working	Significant	High
Producers adding wrong product details	High	Significant
Incorrect stock and availability	Significant	High
Producer struggling to adopt using the dashboard	High	High

Prioritised Risks Mitigation

Risk Description	**Mitigation Strategy**

Quality issues	Define quality standards early in the project and performing quality assurance checks continuously.
Server outages	Set up monitoring tools and implement redundancy systems.
Scope creep	Define clear project scope at the start of the project, while taking account for our technical capability.
Schedule delays	Developing a realistic timeline with buffer time and monitoring progress with regular updates.
Scalability limitation under high traffic	Design for vertical and horizontal scalability and use software that handle scalability automatically.
Ordering system errors	Thorough testing of ordering logic and validation before launch. Admin able to change ordering details from the back end.
Back-end access by unauthorised users	Implementing role-based access and two factor authorisations for users.
Payment provider not working	Monitoring payment provider errors and including other payment providers in case one does not work.
Producers adding wrong product details	Input validation for incorrect numerical data. Edit button to edit details. Confirmation message showing the product details before they are published to the website.
Incorrect stock and availability	Real-time stock updates sent to producers. Automatic stock reduction after an order is approved. Notifications sent to producers when stock is low.
Producer struggling to adopt using the dashboard	Providing the producers with a guidance or tutorial on how to use the website.

Legal Considerations

Legal Regulation	Application
General Data Protection Regulation (GDPR)	This will be done by having a privacy page to tell the customers how their data is protected. Furthermore, cookies and encryption of data will be used to protect the data from being used after a data leak.
Equality Act 2010	To comply with this, the website will have a section to change accessibility settings for the users to fit their needs. For example, text-to-speech, dark mode or font size change.
Payment Services Regulations 2017	To comply, the solution will use external payment providers as they already have secure ways to process payments, which comply with global regulations regarding payment processing. Furthermore, external payment providers will be used since it is industry standard, and they are trusted by many businesses.
Food Information Regulations 2014 and Food Safety Act 1990	This can be done by showing the allergies information on the products page and avoid misleading product descriptions by regularly auditing the products information.

Consumer Rights Act 2015 (CRA)	To comply with this, the product details will need to be confirmed before being published to the solution to ensure the producers check for inaccurate information. Furthermore, there will be a policy page to outline what rights the costumers and producers have, for example right to refund.

UI Considerations

The user interface is going to be simple and minimal, since the users will vary in age juristically, as the customers and producers can be young adults or old people. The colour scheme therefore also will be simple and will evoke a sense of freshness and connection to local farmers. The pictures will be high quality and not busy to match the aesthetics of the whole UI. There will be minimal animations, so customers can easily identify which elements they can interact with and where their attention should go. Any interactable elements will have rounded corners and have animations on interaction, as it is industry standard to do this. To ensure the UI is not busy there will be enough space between different elements that they can be identified as different elements.